

SAFETY DATA SHEET

SECTION 1: IDENTIFICATION

PRODUCT NAME: MOXY TURBO APC
PRODUCT USE: ALL PURPOSE CLEANER
MANUFACTURER/SUPPLIER: MOXY PERFORMANCE
ADDRESS: 104 CHALLENGER CT SANFORD, FL 32771
COMPANY TELEPHONE NUMBER: (866) 524-0359
EMERGENCY RESPONSE NUMBER: 1-800-535-5053 (NORTH AMERICA)
+1-352-323-3500 (INTERNATIONAL)

SECTION 2: HAZARD(S) IDENTIFICATION

***GHS CLASSIFICATION:
HEALTH ENVIRONMENTAL***

PRODUCT DEFINITION: MIXTURE

CLASSIFICATION IN ACCORDANCE WITH 29 CFR 1910
(OSHA HCS) AND REGULATION EC NO. 1272/2008
SKIN CORROSION - CATEGORY 1C [H314]
SINGLE TARGET ORGAN TOXICITY, SINGLE EXPOSURE -
CATEGORY 3; STOT SE 3 [H335]

HAZARD STATEMENTS

DANGER!
H314 - CAUSES SEVERE SKIN BURNS AND EYE DAM-
AGE
H335 - MAY CAUSE RESPIRATORY IRRITATION

PICTOGRAM



GHS05

MOXY

PRECAUTIONARY STATEMENTS

PREVENTION:

P260 – DO NOT BREATHE MIST, SPRAY, OR VAPOR.

P264 – WASH HANDS AND OTHER EXPOSED SKIN AREAS THOROUGHLY AFTER HANDLING.

P271 – USE ONLY OUTDOORS OR IN A WELL-VENTILATED AREA.

P280 – WEAR PROTECTIVE GLOVES, PROTECTIVE CLOTHING, EYE PROTECTION, AND FACE PROTECTION.

RESPONSE:

P301 + P330 + P331 + P310 – IF SWALLOWED: RINSE MOUTH. DO NOT INDUCE VOMITING. IMMEDIATELY CALL A POISON CENTER OR DOCTOR.

P303 + P361 + P353 – IF ON SKIN: REMOVE IMMEDIATELY ALL CONTAMINATED CLOTHING. RINSE SKIN WITH WATER OR SHOWER.

P304 + P340 + P310 – IF INHALED: REMOVE VICTIM TO FRESH AIR AND KEEP AT REST IN A POSITION COMFORTABLE FOR BREATHING. IMMEDIATELY CALL A POISON CENTER OR DOCTOR.

PRECAUTIONARY STATEMENTS CONTI.

P321 – SPECIFIC TREATMENT: CONTACT A POISON CENTER OR DOCTOR. REFER TO SECTION 4 OF THIS SDS.

P363 – WASH CONTAMINATED CLOTHING BEFORE REUSE.

STORAGE AND DISPOSAL STATEMENTS:

P405 – STORE LOCKED UP IN A WELL-VENTILATED PLACE. KEEP CONTAINER TIGHTLY CLOSED.

P501 – DISPOSE OF CONTENTS AND CONTAINERS IN ACCORDANCE WITH NATIONAL AND LOCAL REGULATIONS.

NO OTHER HAZARDS AS DEFINED UNDER 29 CFR 1910.1200.

SECTION 3: COMPOSITION / INFORMATION ON INGREDIENTS

COMPONENT	CAS NUMBER	% WEIGHT	EC NUMBER	INDEX NUMBER
2-BUTOXYETHANOL	111-76-2	≤ 8%	203-905-0	603-014-00-0
SODIUM METASILICATE	6834-92-0	≤ 5%	229-912-9	014-010-00-8
TETRASODIUM EDTA	64-02-8	≤ 5%	200-573-9	
SODIUM HYDROXIDE	1310-73-2	≤ 5%	215-185-5	011-002-00-6
NONYLPHENOL, ETHOZYLATE	127087-87-0	≤ 5%	500-315-8	
LINEAR ALKYL BENZENE SULFONIC ACID	68584-22-5	≤ 5%	271-528-9	

COMPONENT	GHS CLASSIFICATION
2-BUTOXYETHANOL	H227, H302, H312, H315, H319, H332
SODIUM METASILICATE	H290, H314, H335
TETRASODIUM EDTA	H302, H318
SODIUM HYDROXIDE	H290, H314
NONYLPHENOL, ETHOZYLATE	H302, H319, H412
LINEAR ALKYL BENZENE SULFONIC ACID	H319

SECTION 4: FIRST AID MEASURES

INHALATION: IF PRODUCT MIST OR VAPOR CAUSES RESPIRATORY IRRITATION OR DISTRESS, MOVE THE EXPOSED PERSON TO FRESH AIR IMMEDIATELY. IF BREATHING IS DIFFICULT OR IRREGULAR, ADMINISTER OXYGEN; IF RESPIRATORY ARREST OCCURS, START ARTIFICIAL RESPIRATION BY TRAINED PERSONNEL. IF UNCONSCIOUS, MAINTAIN AN OPEN AIRWAY. LOOSEN TIGHT CLOTHING SUCH AS A COLLAR, TIE, BELT, OR WAISTBAND. IF SYMPTOMS PERSIST OR IF THE VICTIM FEELS UNWELL, SEEK MEDICAL ATTENTION.

EYES: IMMEDIATELY FLUSH EYES WITH LARGE AMOUNTS OF WATER OR SALINE SOLUTION FOR AT LEAST 15 MINUTES, OCCASIONALLY LIFTING THE UPPER AND LOWER LIDS. REMOVE CONTACT LENSES, IF PRESENT AND EASY TO DO, AFTER 1-2 MINUTES AND CONTINUE RINSING. SEEK IMMEDIATE MEDICAL ATTENTION, PREFERABLY FROM AN OPHTHALMOLOGIST.

SKIN: FLUSH SKIN WITH LARGE AMOUNTS OF WATER WHILE REMOVING CONTAMINATED CLOTHING. WASH THE AFFECTED AREA WITH SOAP AND WATER FOLLOWED BY THOROUGH RINSING. WASH CONTAMINATED CLOTHING AND SHOES BEFORE REUSE. SEEK IMMEDIATE MEDICAL ATTENTION FOR CHEMICAL BURNS. IF IRRITATION PERSISTS OR IF THE VICTIM FEELS UNWELL, SEEK MEDICAL ATTENTION.

INGESTION: RINSE MOUTH WITH WATER IF THE VICTIM IS CONSCIOUS. REMOVE DENTURES IF PRESENT. GIVE 2 GLASSES OF WATER (MAXIMUM) IF THE VICTIM IS CONSCIOUS, ALERT, AND ABLE TO SWALLOW. DO NOT INDUCE VOMITING UNLESS DIRECTED TO DO SO BY MEDICAL PERSONNEL. VOMITING MAY OCCUR SPONTANEOUSLY. TO PREVENT ASPIRATION OF MATERIAL INTO THE LUNGS, LAY THE VICTIM ON ONE SIDE WITH THE HEAD LOWER THAN THE WAIST. NEVER GIVE ANYTHING BY MOUTH TO AN UNCONSCIOUS OR CONVULSING PERSON. DO NOT LEAVE THE VICTIM UNATTENDED. SEEK IMMEDIATE MEDICAL ATTENTION.

MOST IMPORTANT SYMPTOMS AND EFFECTS, BOTH ACUTE AND DELAYED

POTENTIAL HEALTH SYMPTOMS AND EFFECTS

EYES: CAUSES SEVERE BURNS AND EYE DAMAGE. CONTACT MAY CAUSE ULCERATION OF THE CONJUNCTIVA AND CORNEA. MAY CAUSE IRREVERSIBLE EYE INJURY. MAY CAUSE BLINDNESS. VAPOR OF MIST CAN CAUSE SEVERE EYE IRRITATION. EFFECTS MAY BE DELAYED.

SKIN: CAUSES SKIN IRRITATION WITH LOCALIZED REDNESS, ITCHING AND DISCOMFORT. PROLONGED CONTACT WITH UNPROTECTED SKIN MAY CAUSE DEFATTING OF THE SKIN AND/OR DERMATITIS. MAY BE HARMFUL IF ABSORBED THROUGH THE SKIN.

INHALATION: HARMFUL IF INHALED. INHALATION OF MIST OR VAPOR MAY CAUSE CHEMICAL BURNS TO THE RESPIRATORY TRACT. CAUSES SEVERE IRRITATION WITH COUGH, WHEEZING, LARYNGITIS, BREATHING DIFFICULTY, HEADACHE, AND NAUSEA. THIS MATERIAL IS EXTREMELY DESTRUCTIVE TO TISSUE OF THE MUCOUS MEMBRANES AND UPPER RESPIRATORY TRACT. INHALATION MAY LEAD TO CHEMICAL PNEUMONITIS AND PULMONARY EDEMA. MAY CAUSE INFLAMMATION AND EDEMA OF THE LARYNX AND BRONCHI.

INGESTION: HARMFUL IF SWALLOWED. CAUSES SEVERE BURNS TO THE LIPS, MOUTH, THROAT, AND GASTRO-INTESTINAL TRACT WITH ABDOMINAL PAIN, VOMITING, DIARRHEA, AND POSSIBLE DEATH. MAY CAUSE PERFORATION OF AND SEVERE AND PERMANENT DAMAGE TO THE DIGESTIVE TRACT. MAY CAUSE CIRCULATORY SYSTEM FAILURE.

CHRONIC: INDIVIDUALS WITH PRE-EXISTING SKIN CONDITIONS AND RESPIRATORY DISORDERS MAY BE MORE SUSCEPTIBLE TO THE EFFECTS OF THIS PRODUCT. PROLONGED OR REPEATED SKIN CONTACT MAY CAUSE DRYING AND CRACKING OF THE SKIN, DERMATITIS OR AGGRAVATE EXISTING SKIN CONDITIONS. CHRONIC EYE CONTACT MAY CAUSE CONJUNCTIVITIS AND PERMANENT EYE DAMAGE. CHRONIC EXPOSURE MAY CAUSE DAMAGE TO THE LIVER AND KIDNEYS. 2-BUTOXYETHANOL IS A KNOWN ANIMAL CARCINOGEN. REFER TO SECTION 11.2.

ADVICE TO DOCTOR AND HOSPITAL PERSONNEL

TREAT SYMPTOMATICALLY AND SUPPORTIVELY. TREAT AS FOR STRONG ALKALIS.

SECTION 5: FIRE FIGHTING MEASURES

SUITABLE METHODS OF EXTINCTION: USE EXTINGUISHING MEDIA SUITABLE FOR THE SURROUNDING FIRE.

UNSUITABLE METHODS OF EXTINCTION: NO LIMITATIONS OF EXTINGUISHING AGENTS ARE GIVEN FOR THIS MATERIAL.

SPECIAL HAZARDS ARISING FROM THE SUBSTANCE OR MIXTURE

CLOSED CONTAINERS MAY EXPLODE DUE TO THE BUILDUP OF PRESSURE WHEN EXPOSED TO EXTREME HEAT. DURING EMERGENCY CONDITIONS OVEREXPOSURE TO DECOMPOSITION PRODUCTS MAY CAUSE A HEALTH HAZARD. SYMPTOMS MAY NOT BE IMMEDIATELY APPARENT OR MAY BE DELAYED. OBTAIN MEDICAL ATTENTION.

EXPLOSION HAZARDS: THIS PRODUCT IS NOT CONSIDERED AN EXPLOSION HAZARD.

ADVICE FOR FIREFIGHTERS

FULL PROTECTIVE EQUIPMENT INCLUDING SELF-CONTAINED BREATHING APPARATUS SHOULD BE USED. WATER MAY BE USED TO COOL CLOSED CONTAINERS TO PREVENT PRESSURE BUILDUP AND POSSIBLE AUTO-IGNITION OR EXPLOSION WHEN EXPOSED TO EXTREME HEAT. WATER CONTAMINATED BY THIS MATERIAL MUST BE CONTAINED AND PREVENTED



FROM BEING DISCHARGED TO ANY WATERWAY, SEWER, OR DRAIN.

SECTION 6: ACCIDENT RELEASE MEASURES

EVACUATE NON-ESSENTIAL PERSONNEL. WEAR APPROPRIATE PROTECTIVE CLOTHING AND EQUIPMENT DESIGNATED IN SECTION 8.2. VENTILATE THE AREA. REMOVE ALL SOURCES OF IGNITION. NO SMOKING. CLEAN UP SPILLS IMMEDIATELY. SPILL CREATES A SLIP HAZARD.

ENVIRONMENTAL PRECAUTIONS

AVOID DISPOSAL OF SPILLED MATERIAL TO RUNOFF AND PREVENT CONTACT WITH SOIL AND ENTRY INTO DRAINS, SEWERS, OR WATERWAYS.

METHODS AND MATERIALS FOR CONTAINMENT AND CLEANING UP

APPROACH SPILL FROM UPWIND DIRECTION. DO NOT FLUSH SPILLS DOWN THE DRAIN. COVER DRAINS AND CONTAIN SPILL. COVER SPILL WITH A LARGE QUANTITY OF INERT ABSORBENT. DO NOT USE COMBUSTIBLE MATERIAL SUCH AS SAWDUST. COLLECT MATERIAL AND PLACE INTO AN APPROVED CONTAINER FOR PROPER DISPOSAL. OBSERVE POSSIBLE MATERIAL RESTRICTIONS (SECTIONS 7.2 AND 10.5). DO NOT ALLOW MATERIAL OR RUNOFF FROM RINSING CONTAMINATED AREAS TO ENTER FLOOR DRAINS OR STORM DRAINS AND DITCHES THAT LEAD TO WATERWAYS. DISPOSE OF CONTENTS AND CONTAINERS VIA A LICENSED WASTE DISPOSAL CONTRACTOR.

SECTION 7: HANDLING AND STORAGE

PRECAUTIONS FOR SAFE HANDLING:

WEAR ALL APPROPRIATE PERSONAL PROTECTIVE EQUIPMENT SPECIFIED IN SECTION 8.2. DO NOT GET IN EYES OR ON SKIN OR CLOTHING. DO NOT INHALE MIST OR VAPOR. NO SMOKING. IF NORMAL USE OF MATERIAL PRESENTS A RESPIRATORY HAZARD, USE ONLY ADEQUATE VENTILATION OR WEAR AN APPROPRIATE RESPIRATOR.

WASH CONTAMINATED CLOTHING AND SHOES THOROUGHLY BEFORE REUSE.

ADVICE ON PROTECTION AGAINST FIRE AND EXPLOSION:

THIS PRODUCT IS NOT CONSIDERED TO BE A FIRE OR EXPLOSION HAZARD.

CONDITIONS FOR SAFE STORAGE, INCLUDING ANY INCOMPATIBILITIES:

STORE IN DRY, COOL, WELL-VENTILATED AREAS AWAY FROM INCOMPATIBLE MATERIALS (SEE SECTION 10.5), FOOD, AND DRINK. KEEP AWAY FROM STRONG ACIDS. KEEP FROM FREEZING. TRANSFER ONLY TO APPROVED CONTAINERS HAVING CORRECT LABELING. KEEP CONTAINERS TIGHTLY CLOSED WHEN NOT IN USE. PROTECT CONTAINERS AGAINST PHYSICAL DAMAGE. CONTAINERS THAT HAVE BEEN OPENED MUST BE CAREFULLY RESEALED AND KEPT UPRIGHT TO PREVENT LEAKAGE. CONTAINERS ARE HAZARDOUS WHEN EMPTY AS THEY CONTAIN PRODUCT RESIDUE. USE APPROPRIATE CONTAINMENT TO AVOID ENVIRONMENTAL CONTAMINATION. VENTILATE CLOSED AREAS. KEEP LOCKED UP AND OUT OF REACH OF CHILDREN. 7.3

SPECIFIC END USES:

APART FROM THE USES MENTIONED IN SECTION 1.2, NO OTHER SPECIFIC USES ARE STIPULATED.

SECTION 8: EXPOSURE CONTROL/PERSONAL PROTECTION

CONTROL PARAMETERS:

OCCUPATIONAL EXPOSURE LIMIT VALUES

CAS NUMBER	INGREDIENT	OSHA PEL	ACGIH TLV	NIOSH
111-76-2	2-Butoxyethanol	50 ppm; 240 mg/m ³ TWA	20 ppm; 97 mg/m ³ TWA; skin	50 ppm; 24 mg/m ³ TWA 700 ppm IDLH; skin
1310-73-2	Sodium hydroxide	2mg/m ³ TWA	2mg/m ³ , ceiling	2mg/m ³ , ceiling; 10mg/m ³ IDHL

A “skin” notation following the inhalation exposure guideline refers to the potential for dermal absorption of the material, including eyes and mucous membranes, either by direct contact with vapors or by direct skin contact. It is intended to alert the reader that inhalation may not be the only route of exposure and that measures to minimize dermal exposure should be considered

EXPOSURE CONTROLS:

ENGINEERING MEASURES: TECHNICAL MEASURES AND APPROPRIATE WORKING OPERATIONS SHOULD BE GIVEN PRIORITY OVER THE USE OF PERSONAL PROTECTIVE EQUIPMENT. USE ADEQUATE VENTILATION. LOCAL EXHAUST IS PREFERABLE. REFER TO SECTION 7.1.

INDIVIDUAL PROTECTION MEASURES: WEAR PROTECTIVE CLOTHING TO PREVENT REPEATED OR PROLONGED CONTACT WITH PRODUCTS. PROTECTIVE CLOTHING NEEDS TO BE SELECTED SPECIFICALLY FOR THE WORK-PLACE, DEPENDING ON CONCENTRATIONS AND QUANTITIES OF HAZARDOUS SUBSTANCES HANDLED. THE CHEMICAL RESISTANCE OF THE PROTECTIVE EQUIPMENT SHOULD BE ENQUIRED AT THE REPRESENTATIVE SUPPLIER.

HYGIENE MEASURES: FACILITIES STORING OR USING THIS MATERIAL SHOULD BE EQUIPPED WITH AN EYE-WASH STATION AND SAFETY SHOWER. CHANGE CONTAMINATED CLOTHING. PREVENTIVE SKIN PROTECTION IS RECOMMENDED. WASH HANDS THOROUGHLY AFTER USE, BEFORE EATING, DRINKING, SMOKING, OR USING THE LAVATORY.

EYE/FACE PROTECTION: WEAR SAFETY GLASSES WITH UNPERFORATED SIDE SHIELDS OR PROTECTIVE SPLASH GOGGLES DURING USE.

HAND PROTECTION: WEAR GLOVES RECOMMENDED BY GLOVE SUPPLIER FOR PROTECTION AGAINST MATERIALS IN SECTION 3. GLOVES SHOULD BE IMPERMEABLE TO CHEMICALS AND OIL. BREAKTHROUGH TIME OF SELECTED GLOVES MUST BE GREATER THAN THE INTENDED USE PERIOD.

SKIN PROTECTION: WEAR PROTECTIVE CLOTHING. WEAR PROTECTIVE BOOTS IF THE SITUATION REQUIRES.

RESPIRATORY PROTECTION: ALWAYS USE AN APPROVED RESPIRATOR WHEN VAPOR/AEROSOLS EXCEED PERMISSIBLE EXPOSURE LIMITS. WHERE RISK ASSESSMENT SHOWS AIR-PURIFYING RESPIRATORS ARE APPROPRIATE, USE A HALF-MASK RESPIRATOR WITH MULTI-PURPOSE COMBINATION (US) OR TYPE ABEK (EN 14387) RESPIRATOR CARTRIDGES AS A BACKUP TO ENGINEERING CONTROLS. IF THE RESPIRATOR IS THE SOLE MEANS OF PROTECTION, USE A FULL-FACE SUPPLIED AIR RESPIRATOR. USE RESPIRATORS AND COMPONENTS TESTED AND APPROVED UNDER APPROPRIATE GOVERNMENT STANDARDS SUCH AS NIOSH (US) OR CEN (EU). FOLLOW OSHA RESPIRATOR REGULATIONS FOUND IN 29 CFR 1910.134 OR EUROPEAN STANDARD EN 149

ENVIRONMENTAL EXPOSURE CONTROLS: DO NOT EMPTY INTO DRAINS.

PPE must not be considered a long-term solution to exposure control. PPE usage must be accompanied by employer programs to properly select, maintain, clean, fit, and use. Consult a competent industrial hygiene resource to determine hazard potential and/or the PPE manufacturers to ensure adequate protection



SECTION 9: PHYSICAL AND CHEMICAL PROPERTIES**INFORMATION ON BASIC PHYSICAL AND CHEMICAL PROPERTIES**

APPEARANCE:	CLEAR, YELLOW LIQUID
ODOR	CHARACTERISTIC
ODOR THRESHOLD	NO DATA AVAILABLE
MOLECULAR WEIGHT	NO DATA AVAILABLE
CHEMICAL FORMULA	NO DATA AVAILABLE
PH	12.0-14.0
FREEZING/MELTING POINT	NO DATA AVAILABLE
INITIAL BOILING POINT	100 °C(212 °F)
EVAPORATION RATE	NO DATA AVAILABLE
FLAMMABILITY (SOLID, GAS)	NOT APPLICABLE
FLASH POINT	NOT APPLICABLE
AUTOIGNITION TEMPERATURE	NO DATA AVAILABLE
DECOMPOSITION TEMPERATURE	NO DATA AVAILABLE
LOWER EXPLOSIVE LIMIT	(LEL)NO DATA AVAILABLE
UPPER EXPLOSIVE LIMIT	(UEL)NO DATA AVAILABLE
VAPOR PRESSURE	NO DATA AVAILABLE
VAPOR DENSITY	NO DATA AVAILABLE
DENSITY	0.994-1.034 G/ML (8.29-8.63LB/GAL)
VISCOSITY	NO DATA AVAILABLE
SOLUBILITY IN WATER	COMPLETELY MISCIBLE
PARTITION COEFFICIENT (N-OCTANOL/WATER)	NO DATA AVAILABLE
OXIDIZING PROPERTIES	NOT APPLICABLE
EXPLOSIVE PROPERTIES	NOT APPLICABLE
VOLATILES BY WEIGHT @ 21 °C	93.45%
VOC(WT.%)	3.5% (35 G/ML; 0.29 LB/GAL)

SECTION 10: STABILITY AND REACTIVITY**REACTIVITY**

NO SPECIAL REACTIVITY HAS BEEN REPORTED DURING NORMAL CONDITIONS OF HANDLING AND USE.

CHEMICAL STABILITY

THIS MATERIAL IS STABLE UNDER RECOMMENDED STORAGE AND HANDLING CONDITIONS.

POSSIBILITY OF HAZARDOUS REACTIONS

HAZARDOUS POLYMERIZATION WILL NOT OCCUR.

CONDITIONS TO AVOID

AVOID TEMPERATURE EXTREMES AND CONTACT WITH INCOMPATIBLE MATERIALS.

INCOMPATIBLE MATERIALS

STRONG OXIDIZING AGENTS, ACIDS

HAZARDOUS DECOMPOSITION PRODUCTS

THERMAL DECOMPOSITION PRODUCTS MAY INCLUDE OXIDES OF CARBON,AND SODIUM OXIDE FUME

SECTION II: TOXICOLOGICAL INFORMATION**11.1 INFORMATION ON TOXICOLOGICAL EFFECTS**

ACUTE ORAL TOXICITY	NO DATA AVAILABLE
ACUTE INHALATION TOXICITY	NO DATA AVAILABLE
ACUTE DERMAL TOXICITY	NO DATA AVAILABLE
SKIN IRRITATION	CAUSES SEVERE SKIN BURNS.
EYE IRRITATION	CAUSES SERIOUS EYE DAMAGE
SENSITIZATION	NO DATA AVAILABLE
CARCINOGENICITY	NO DATA AVAILABLE
GERM CELL MUTAGENICITY	NO DATA AVAILABLE
REPRODUCTIVE TOXICITY	NO DATA AVAILABLE
SPECIFIC ORGAN TOXICITY-SINGLE EXPOSURE	MAY CAUSE RESPIRATORY IRRITATION.
SPECIFIC ORGAN TOXICITY-REPEATED EXPOSURE	NO DATA AVAILABLE
ASPIRATION HAZARD	NO DATA AVAILABLE

11.2 FURTHER INFORMATION

2-BUTOXYETHANOL(CAS #111-76-2): IARC GROUP 3 CARCINOGEN-NOT CLASSIFIABLE AS TO ITS CARCINOGENICITY TO HUMANS. NOT LISTED AS A CARCINOGEN BY ACGIH, NTP OR OSHA. IN LONG-TERM ANIMAL STUDIES WITH 2-BUTOXYETHANOL, SMALL BUT STATISTICALLY SIGNIFICANT INCREASES IN TUMORS WERE OBSERVED IN MICE BUT NOT RATS. THE EFFECTS ARE NOT BELIEVED TO BE RELEVANT TO HUMANS.

2-BUTOXYETHANOL INHALATION EXPOSURE IN LABORATORY ANIMALS HAS BEEN FOUND TO REDUCE BODY WEIGHT GAIN AND FOOD CONSUMPTION IN ADDITION TO HEMOLYSIS. AFTER EXPOSURE WAS DISCONTINUED, THESE EFFECTS IN ANIMALS DISAPPEARED. ADVERSE REPRODUCTIVE OR BIRTH EFFECTS WERE NOT REPORTED IN ANIMALS EXCEPT WHEN EXPOSURES WERE HIGH ENOUGH TO CAUSE SIGNIFICANT MATERNAL TOXICITY. IN ANIMALS, HEMOLYSIS (RED BLOOD CELL BREAKAGE) AND SECONDARY EFFECTS TO THE KIDNEYS AND LIVER HAVE BEEN REPORTED. HUMAN RED BLOOD CELLS HAVE BEEN SHOWN TO BE SIGNIFICANTLY LESS SENSITIVE TO HEMOLYSIS THAN THOSE OF RODENTS AND RABBITS.

NO DATA IS AVAILABLE REGARDING THE MUTAGENICITY OR TERATOGENICITY OF THIS PRODUCT, NOR IS THERE ANY AVAILABLE DATA THAT INDICATES THAT IT CAUSES ADVERSE DEVELOPMENTAL OR FERTILITY EFFECTS.

HANDLE IN ACCORDANCE WITH GOOD INDUSTRIAL HYGIENE AND SAFETY PRACTICE

SECTION 12: ECOLOGICAL INFORMATION

TOXICITY

LARGE SPILLS OR DISCHARGES OF THIS PRODUCT MAY BE HARMFUL TO AQUATIC LIFE. LARGE SPILLS OR DISCHARGES MAY INCREASE THE PH OF AQUATIC SYSTEMS AND CAN BE HARMFUL TO AQUATIC LIFE AND SOIL MICROORGANISMS. 2-BUTOXYETHANOL IS HARMFUL TO ALGAE AND HIGHER AQUATIC PLANTS.

PERSISTENCE AND DEGRADABILITY

ORGANIC COMPONENTS IN THIS PRODUCT ARE BIODEGRADABLE. INORGANIC SUBSTANCES ARE NOT BIODEGRADABLE. METHODS FOR THE DETERMINATION OF BIODEGRADABILITY ARE NOT APPLICABLE TO INORGANIC SUBSTANCES.

BIOACCUMULATION POTENTIAL

LOW

MOBILITY IN SOIL

NO DATA AVAILABLE

RESULTS OF PBT AND VPVB ASSESSMENT

NO DATA AVAILABLE

OTHER EFFECTS

ADDITIONAL ECOLOGICAL INFORMATION:

DO NOT ALLOW MATERIAL TO RUN INTO SURFACE WATERS, WASTEWATER, OR SOIL.

AN ENVIRONMENTAL HAZARD CANNOT BE EXCLUDED IN THE EVENT OF UNPROFESSIONAL HANDLING OR DISPOSAL

SECTION 13: DISPOSAL CONSIDERATIONS

WASTE TREATMENT METHODS

METHODS OF DISPOSAL: THE GENERATION OF WASTE SHOULD BE AVOIDED OR MINIMIZED WHENEVER POSSIBLE. EMPTY CONTAINERS OR LINERS MAY RETAIN SOME PRODUCT RESIDUES. THIS MATERIAL AND ITS CONTAINER MUST BE DISPOSED OF IN A SAFE WAY. DISPOSE OF SURPLUS AND NON-RECYCLABLE PRODUCTS IN ACCORDANCE WITH NATIONAL, STATE AND LOCAL REGULATIONS. DISPOSAL OF THIS PRODUCT, SOLUTIONS, AND ANY BY-PRODUCTS SHOULD AT ALL TIMES COMPLY WITH THE REQUIREMENTS OF ENVIRONMENTAL PROTECTION AND WASTE DISPOSAL LEGISLATION AND ANY REGIONAL LOCAL AUTHORITY REQUIREMENTS. AVOID DISPERSAL OF SPILT MATERIAL AND RUNOFF AND CONTACT WITH SOIL, WATERWAYS, DRAINS, AND SEWERS.

RCRA F-SERIES: NO LISTINGS ABOVE THE REPORTABLE THRESHOLD (DE MINIMIS)

RCRA U-SERIES: NO LISTINGS ABOVE THE REPORTABLE THRESHOLD (DE MINIMIS)

SECTION 14: TRANSPORT INFORMATION

NOTE:TRANSPORTATION INFORMATION PROVIDED IS FOR REFERENCE ONLY. CUSTOMER IS URGED TO CONSULT 49 CFR 100-177, IMDG, IATA, EC, UNITED NATIONS TDG,AND WHMIS (CANADA) TDG INFORMATION MANUALS FOR DETAILED REGULATIONS AND EXCEPTIONS COVERING SPECIFIC CONTAINER SIZES, PACKAGING MATERIALS, AND METHODS OF SHIPPING.

LIMITED QUANTITY FOR CORROSIVE LIQUIDS PACKING GROUP III WHEN INNER PACKAGINGS ARE NOT OVER 5.0 LITERS(1.3 GALLONS) NET CAPACITY EACH, PACKED IN STRONG OUTER PACKAGING.

USA DOT (GROUND TRANSPORTATION)

PROPER SHIPPING NAME	CORROSIVE LIQUIDS, N.O.S. (SODIUM HYDROXIDE; SODIUM METASILICATE)
HAZARD CLASS	8
UN	UN1760
PACKING GROUP	III
NAERG	GUIDE #154
PACKAGING AUTHORIZATION	NON-BULK:49 CFR 173.203; BULK:173.241
PACKAGING EXCEPTIONS	49 CFR 173.154

IMO/IMDG (WATER TRANSPORTATION)

PROPER SHIPPING NAME	CORROSIVE LIQUIDS, N.O.S. (SODIUM HYDROXIDE; SODIUM METASILICATE)
HAZARD CLASS	8
UN	UN1760
PACKING GROUP	III
MARINE POLLUTANT	NO
EMS NUMBER	F-A, S-B

ICAO/IATA (AIR TRANSPORTATION)

PROPER SHIPPING NAME	CORROSIVE LIQUIDS, N.O.S. (SODIUM HYDROXIDE; SODIUM METASILICATE)
HAZARD CLASS	8
UN	UN1760
PACKING GROUP	III
QUANTITY LIMITATIONS	49 CFR 175.27 AND 175.75-CARGO AIRCRAFT ONLY:60 L; PASSENGER AIRCRAFT:5L

RID/ADR (RAIL TRANSPORTATION)

PROPER SHIPPING NAME	CORROSIVE LIQUIDS, N.O.S. (SODIUM HYDROXIDE; SODIUM METASILICATE)
HAZARD CLASS	8
UN	UN1760
PACKING GROUP	III



SECTION 15: REGULATORY INFORMATION**SAFETY, HEALTH AND ENVIRONMENTAL REGULATIONS/LEGISLATION SPECIFIC FOR SUBSTANCE OR MIXTURE****U.S. FEDERAL REGULATIONS**

OSHA HAZARD COMMUNICATION STANDARD: THIS MATERIAL IS CLASSIFIED AS HAZARDOUS IN ACCORDANCE WITH OSHA 29 CFR 1910.1200.

TOXIC SUBSTANCE CONTROL ACT (TSCA) INVENTORY: ALL SUBSTANCES IN THIS PRODUCT ARE LISTED ON THE TSCA INVENTORY. THIS PRODUCT IS NOT SUBJECT TO TSCA 12(B) EXPORT NOTIFICATION.

OSHA PROCESS SAFETY MANAGEMENT STANDARD: THIS PRODUCT IS NOT REGULATED UNDER OSHA PSM STANDARD 29 CFR 1910.119.

EPA RISK MANAGEMENT PLANNING STANDARD: THIS PRODUCT IS NOT REGULATED UNDER EPA RMP STANDARD (RMP) 40 CFR PART 68.

EPA FEDERAL INSECTICIDE, FUNGICIDE AND RODENTICIDE ACT: THIS PRODUCT IS NOT A REGISTERED PESTICIDE UNDER THE FIFRA, 40 CFR PART 150.

TOXIC SUBSTANCE CONTROL ACT (TSCA) INVENTORY: ALL SUBSTANCES IN THIS PRODUCT ARE LISTED ON THE TSCA INVENTORY. THIS PRODUCT IS NOT SUBJECT TO TSCA 12(B) EXPORT NOTIFICATION.

DRUG ENFORCEMENT ADMINISTRATION (DEA) LIST 2, ESSENTIAL CHEMICALS (21 CFR 1310.02(B)) AND 1310.4(F)(2)) AND CHEMICAL CODE NUMBER: NO LISTINGS.

DRUG ENFORCEMENT ADMINISTRATION (DEA) LISTS 1 & 2, EXEMPT CHEMICAL MIXTURES (21 CFR 1310.12(C)) AND CODE NUMBER: NO LISTINGS.

DEPARTMENT OF HOMELAND SECURITY (DHS) CHEMICAL FACILITY ANTI-TERRORISM STANDARDS (CFATS) CHEMICALS: NO LISTINGS.

SUPERFUND AMENDMENTS AND REAUTHORIZATION ACT (SARA)

SARA SECTION 311/312 HAZARD CATEGORIES: CAUSES SEVERE SKIN BURNS AND EYE DAMAGE. MAY CAUSE RESPIRATORY IRRITATION

SARA 313 INFORMATION: GLYCOL ETHERS (CAS NUMBER N230) ARE SUBJECT TO THE REPORTING LEVELS ESTABLISHED BY SECTION 313 OF THE EMERGENCY PLANNING AND COMMUNITY RIGHT-TO-KNOW ACT OF 1986.

SARA 302/304 EXTREMELY HAZARDOUS SUBSTANCE: NONE OF THE COMPONENTS OF THIS PRODUCT EXCEED THE THRESHOLD (DE MINIMIS) REPORTING LEVELS ESTABLISHED BY THESE SECTIONS OF TITLE III OF SARA.

SARA 302/304 EMERGENCY PLANNING AND NOTIFICATION: NONE OF THE COMPONENTS OF THIS PRODUCT EXCEED THE THRESHOLD (DE MINIMIS) REPORTING LEVELS ESTABLISHED BY THESE SECTIONS OF TITLE III OF SARA.

COMPREHENSIVE RESPONSE COMPENSATION AND LIABILITY ACT (CERCLA): THIS PRODUCT CONTAINS THE FOLLOWING CERCLA REPORTABLE SUBSTANCES:

SODIUM HYDROXIDE (CAS #1310-73-2): RQ = 454 KG (1,000 LB)

GLYCOL ETHERS: THERE IS NO RQ ASSIGNED TO THIS BROAD CLASS, ALTHOUGH THE CLASS IS A CERCLA HAZARDOUS SUBSTANCES. REFER TO 50 FEDERAL REGISTER 13456 (APRIL 4, 1985).

CLEAN AIR ACT (CAA)

THIS PRODUCT DOES NOT CONTAIN HAZARDOUS AIR POLLUTANTS (HAPS) DESIGNATED IN CAA SECTION 112 (B).

THIS PRODUCT DOES NOT CONTAIN CLASS 1 OZONE DEPLETERS.

THIS PRODUCT DOES NOT CONTAIN CLASS 2 OZONE DEPLETERS.

CLEAN WATER ACT (CWA)

GLYCOL ETHERS (EDF-109) AND POTASSIUM HYDROXIDE ARE HAZARDOUS SUBSTANCES.

THIS PRODUCT DOES NOT CONTAIN ANY PRIORITY POLLUTANTS.

THIS PRODUCT DOES NOT CONTAIN ANY TOXIC POLLUTANTS.

U.S. STATE REGULATIONS

CALIFORNIA PROP 65, SAFE DRINKING WATER AND TOXIC ENFORCEMENT ACT OF 1986

THIS PRODUCT CONTAINS NO CHEMICAL(S) KNOWN TO THE STATE OF CALIFORNIA TO CAUSE

CANCER BIRTH DEFECTS OR REPRODUCTIVE HARM IN CONCENTRATIONS THAT EXCEED THE THRESHOLD (DE MINIMIS) REPORTING LEVELS ESTABLISHED UNDER PROPOSITION 65.

OTHER U.S. STATE INVENTORIES

2-BUTOXYETHANOL (CAS #111-76-2) IS LISTED ON THE FOLLOWING STATE HAZARDOUS SUBSTANCE INVENTORIES, RIGHT-TO-KNOW LISTS, AND/OR AIR QUALITY/AIR POLLUTANTS LISTS: CA, MN, PA, RI, WI.

SODIUM HYDROXIDE (CAS #1310-73-2) IS LISTED ON THE FOLLOWING STATE HAZARDOUS SUBSTANCE INVENTORIES, RIGHT-TO-KNOW LISTS, AND/OR AIR QUALITY/AIR POLLUTANTS LISTS: CA, DE, ID, MA, MN, NJ, PA, RI, WA, WI.

CANADA

WHMIS HAZARD CLASSIFICATION: CAUSES SKIN BURNS AND EYE DAMAGE

CANADIAN NATIONAL POLLUTANT RELEASE INVENTORY (NPRI): 2-BUTOXYETHANOL IS LISTED ON THE NPRI.

EUROPEAN ECONOMIC COMMUNITY

GERMANY (WATER DANGER/PROTECTION): 1 (SLIGHTLY HAZARDOUS TO WATER)

Global Chemical Inventory Lists

COUNTRY	INVENTORY NAME	LISTED
Canada	Domestic Substance List (DSL)	Yes
Canada	Non-Domestic Substance List (NDSL)	No
Europe	Inventory of New and Existing Chemicals (EINECS)	Yes
United States	Toxic Substance Control Act (TSCA)	Yes
Australia	Australian Inventory of Chemical Substances (AICS)	Yes
New Zealand	New Zealand Inventory of Chemicals (NZIoC)	Yes
China	Inventory of Existing Chemical Substances in China (IECSC)	Yes
Japan	Inventory of Existing and New Chemical Substances (ENCS)	Yes
Korea	Existing Chemicals List (ECL)	Yes
Philippines	Philippines Inventory of Chemicals and Chemical Substances (PICCS)	Yes

* YES - ALL COMPONENTS OF THIS PRODUCT COMPLY WITH THE INVENTORY REQUIREMENTS ADMINISTERED BY THE GOVERNING COUNTRY.

NO - ONE OR MORE COMPONENTS OF THIS PRODUCT ARE NOT ON THE INVENTORY OR ARE EXEMPT FROM LISTING.

CHEMICAL SAFETY ASSESSMENT: A CHEMICAL SAFETY ASSESSMENT WAS NOT CARRIED OUT FOR THIS PRODUCT.

SECTION 16: OTHER INFORMATION

HMIS HAZARDOUS CLASSIFICATION

HEALTH HAZARD: 3

FLAMMABILITY: 0

INSTABILITY: 0

PERSONAL PROTECTION: C

DISCLAIMER OF RESPONSIBILITY

THE INFORMATION ON THIS SDS WAS OBTAINED FROM SOURCES WHICH WE BELIEVE ARE RELIABLE. HOWEVER, THE INFORMATION IS PROVIDED WITHOUT ANY WARRANTY, EXPRESSED OR IMPLIED, REGARDING ITS CORRECTNESS. SOME INFORMATION PRESENTED, AND CONCLUSIONS DRAWN HEREIN ARE FROM SOURCES OTHER THAN DIRECT TEST DATA ON THE SUBSTANCE ITSELF. THE CONDITIONS OR METHODS OF HANDLING, STORAGE, USE AND DISPOSAL OF THE PRODUCT ARE BEYOND OUR CONTROL AND MAY BE BEYOND OUR KNOWLEDGE. FOR THIS AND OTHER REASONS, WE DO NOT ASSUME DAMAGE OR EXPENSE ARISING OUT OF OR IN ANY WAY RESPONSIBILITY AND EXPRESSLY DISCLAIM LIABILITY FOR LOSS, CONNECTED WITH HANDLING, STORAGE, USE, OR DISPOSAL OF THIS PRODUCT. IF THE PRODUCT IS USED AS A COMPONENT IN ANOTHER PRODUCT, THIS SDS INFORMATION MAY NOT BE APPLICABLE.

REVISION DATE: 08 AUGUST 2023, VERSION 3

SUPERSEDES SDS: 17 FEBRUARY 2021, VERSION 2



Full Text of GHS Hazard Phrases Referenced in Section 3 (not covered in Section 2)

H227 – Combustible liquid
H302 – Harmful if swallowed
H312 – Harmful if inhaled
H314 – Causes skin burns and serious eye damage
H318 – Causes serious eye damage
H332 – Harmful if inhaled
H401 – Toxic to aquatic life
H412 – Harmful to aquatic life with long lasting effects

Abbreviation Key**Abbreviation Full Form**

ACGIH American Conference of Governmental Industrial Hygienists

ADR Accord Dangereux Routier (European regulations concerning the international transport of dangerous goods by road)

CAS Chemical Abstract Services

CFR Code of Federal Regulations

COC Cleveland Open Cup

DOT Department of Transportation

EC50 Half maximal effective concentration

EMS Emergency Response Procedures for Ships Carrying Hazardous Goods

EPA Environmental Protection Agency

ErC50 Reduction of Growth Rate

ERG Emergency Response Guide Book

FDA Food and Drug Administration

GHS Globally Harmonized System of Classification and Labelling of Chemicals (GHS)

HCS Hazard Communication Standard

IARC International Agency for Research on Cancer

LDLo Lowest Lethal Dose

mppcf Millions of Particles Per Cubic Foot

NA North America

NAERG North American Emergency Response Guide Book

NIOSH National Institute for Occupational Safety & Health

NTP National Toxicology Program

OSHA Occupational Safety and Health Administration

PBT Persistent, Bioaccumulating and Toxic

PEL Permissible exposure limit

PMCC Pensky–Martens Closed Cup

ppm Parts Per Million

RCRA Resource Conservation and Recovery Act

RID Dangerous Goods by Rail

RQ Reportable Quantity

TCC/Tag Tagliabue Closed Cup

IATA International Air Transport Association

IC50 Half Maximal Inhibitory Concentration

ICAO International Civil Aviation Organization

IDLH Immediately Dangerous to Life and Health

IMDG International Maritime Dangerous Goods

IMO International Maritime Organization

LC50 50% Lethal Concentration

LD50 50% Lethal Dose

TLV Threshold Limit Value

TSCA Toxic Substance Control Act

TWA Time-weighted Average

UN United Nations

VOC Volatile Organic Compounds

vPvB Very Persistent and Very Bioaccumulating

WHMIS Workplace Hazardous Materials Information System